



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2008

BIOLOGY
HIGHER 2
PAPER 1

9747 / 1
28 AUG 2008

TIME 1 hour 15 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number on the multiple choice answer sheet provided.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

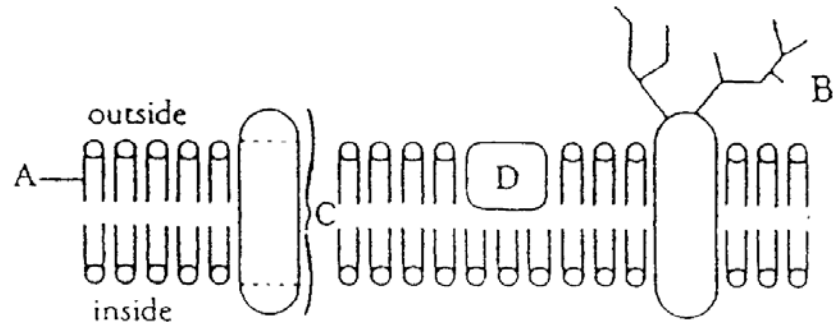
Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 14 printed pages

For Q1-2, refer to the following diagram which shows part of a cell membrane.



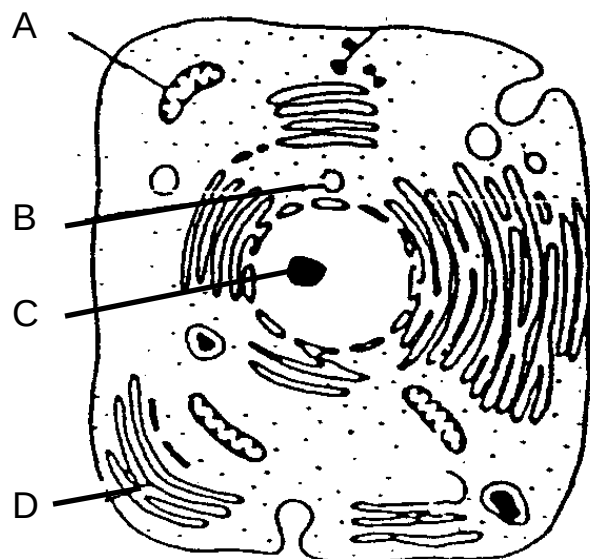
1. Which structure(s) contain(s) the presence of a peptide bond?

- A** A only
- B** B only
- C** A and B
- D** C and D

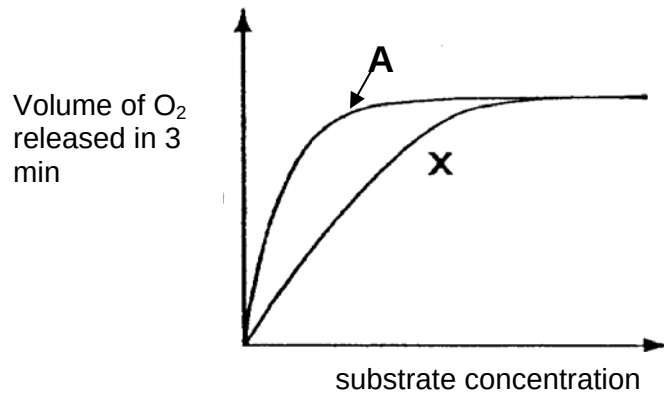
2. Which structure(s) restrict(s) the entry of hydrophilic substances into the cell?

- A** A only
- B** B only
- C** A and C
- D** B and C

3. The diagram below shows an animal cell. Which of the following is responsible for the synthesis of rRNA?

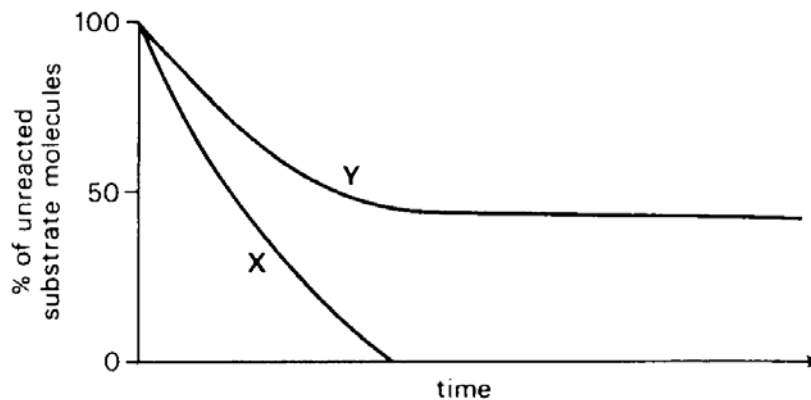


4. Catalase was added to 15ml of hydrogen peroxide at 37°C, and the results were plotted on Graph A. Which of the following would result in graph X?



- A addition of a competitive inhibitor
- B addition of a non-competitive inhibitor
- C addition of less catalase
- D heating to 48°C

5. Graph X shows the results of an enzyme reaction.



Which of the following would result in Graph Y?

- A addition of a non-competitive inhibitor
- B decreased substrate concentration
- C end-product inhibition
- D increased substrate concentration

6. Part of a prokaryotic gene is 228 base pairs long. What is the length of amino acids synthesized by this DNA sequence?

- A 38
- B 76
- C 228
- D 456

7. Which of the following catalyses the formation of a peptide bond during translation?

- A** aminoacyl-tRNA synthetase
- B** DNA polymerase
- C** peptidyl transferase
- D** RNA polymerase

8. The table below shows some amino acids and their DNA genetic code.

Amino acid	DNA code
Alanine	CGG
Lysine	TTT
Arginine	GCG
Phenylalanine	AAA
Glycine	CCA
Valine	CAA

A DNA sequence codes for a polypeptide arginine-glycine-phenylalanine-valine-alanine.

The following mRNA molecule was transcribed from the DNA sequence above. Which mRNA triplet contains a single transcription error?

CGCGUUUUUGUUGCC

- A** the first
- B** the second
- C** the third
- D** the fourth

9. How does DNA synthesis along the lagging strand differ from that on the leading strand?

- A** an RNA primer is needed on the lagging strand, but not on the leading strand.
- B** deoxyribonucleotides are added to the 5' end instead of the 3' end.
- C** helicase synthesizes Okazaki fragments, which are then linked together.
- D** Okazaki fragments, synthesized 5' → 3', are linked by DNA ligase.

10. Red-green colour-blindness is a sex-linked, recessive trait. A colour-blind man marries a woman with normal vision, whose father is colour-blind. If they have a daughter, what is the probability that she will be colour-blind?

- A** 0
- B** 1/4
- C** 1/2
- D** 3/4

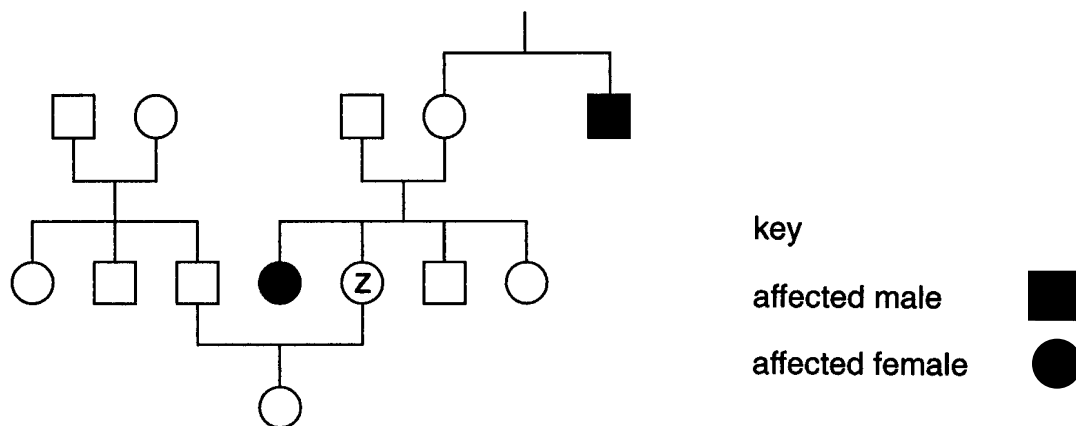
11. In Shorthorn cattle, the allele for hornless is dominant to the allele for the presence of horns. Coat colour can be red (genotype $C^R C^R$), roan (genotype $C^R C^W$) or white (genotype $C^W C^W$).

A roan bull, heterozygous for the hornless trait, is crossed with a cow of the same genotype.

What is the probability that a calf from this cross would have the same genotype as its parents?

- A $1/4$
- B $3/8$
- C $1/2$
- D $3/4$

12. Galactosaemia is a condition caused by a recessive allele of an autosomal gene in which galactose cannot be metabolised. The diagram shows the pattern of inheritance of galactosaemia in a family.



What is the chance that individual Z is a carrier of the condition?

- A $1/2$
- B $1/4$
- C $1/3$
- D $2/3$

13. In the sweet-pea plant, flowers may be either purple ($R_S_$) or white. If two purple plants each of the genotype $RrSs$ are crossed, what is the expected phenotypic ratio of offspring?

- A 3 purple to 1 white
- B 1 purple to 3 white
- C 9 purple to 7 white
- D 13 purple to 3 white

14. A plant is heterozygous for a pair of alleles that are codominant. This plant is self-pollinated and the resulting seeds are germinated and allowed to grow. Which ratios are expected in the offspring?

	ratio of phenotypes	ratio of genotypes
A	1 : 2 : 1	1 : 2 : 1
B	1 : 2 : 1	3 : 1
C	3 : 1	1 : 2 : 1
D	3 : 1	3 : 1

15. A group of small fish live in a lake with a uniformly light-brown sandy bottom. Most of the fish are light brown, but about 10% are mottled. This fish species is often prey for large birds that live on the shore. A construction company dumps a load of gravel at the bottom of the lake, giving it a mottled appearance. Which of these statements presents the most accurate prediction of what will happen to this fish population?

- A** As the mottled fish are eaten, more will be produced to fill the gap.
- B** In two generations, all the fish will be mottled.
- C** The proportion of mottled fish will increase over time.
- D** The ratio of 9 light brown: 1 mottled fish will not change.

16. Which of the following pairs is *least* likely to represent homology?

- A** the haemoglobin of a baboon and that of a gorilla
- B** the mitochondria of a plant and those of an animal
- C** the wings of a bird and those of an insect
- D** the wings of a bat and the arms of a human

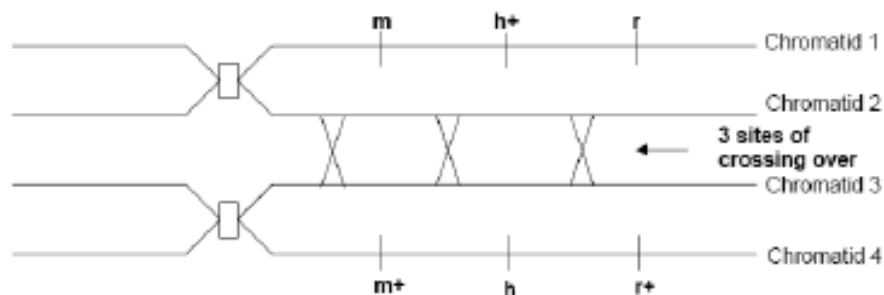
17. Within a few weeks of treatment with the drug 3TC, a patient's HIV population consists entirely of 3TC-resistant viruses. How can this result best be explained?

- A** A few drug-resistant viruses were present at the start of treatment, and natural selection increased their frequency.
- B** HIV began making drug-resistant versions of reverse transcriptase in response to the drug.
- C** The drug caused the HIV single-stranded RNA to change.
- D** The patient must have become re-infected with 3TC-resistant viruses.

18. During a lecture on evolution, one student remarked: "The giraffe stretched its neck while reaching for higher leaves; its offspring inherited longer necks as a result." Which statement is most likely to be helpful in correcting this student's misconception?

A Characteristics acquired during an organism's life are generally not passed on through genes.
B Disuse of an organ may lead to its eventual disappearance.
C Overproduction of offspring leads to a struggle for survival.
D Spontaneous mutations can result in the appearance of new traits.

19. The diagram below shows a pair of homologous chromosomes during prophase of meiosis. Determine the resulting segregation of the named alleles.



A chromatid 1: m h⁺ r
 chromatid 2: m h r
 chromatid 3: m⁺ h⁺ r⁺
 chromatid 4: m⁺ h r⁺
B chromatid 1: m h⁺ r
 chromatid 2: m⁺ h r⁺
 chromatid 3: m h⁺ r
 chromatid 4: m⁺ h r⁺
C chromatid 1: m h⁺ r
 chromatid 2: m⁺ h⁺ r⁺
 chromatid 3: m h r
 chromatid 4: m⁺ h r⁺
D chromatid 1: m h⁺ r
 chromatid 2: m⁺ h⁺ r
 chromatid 3: m h r⁺
 chromatid 4: m⁺ h r⁺

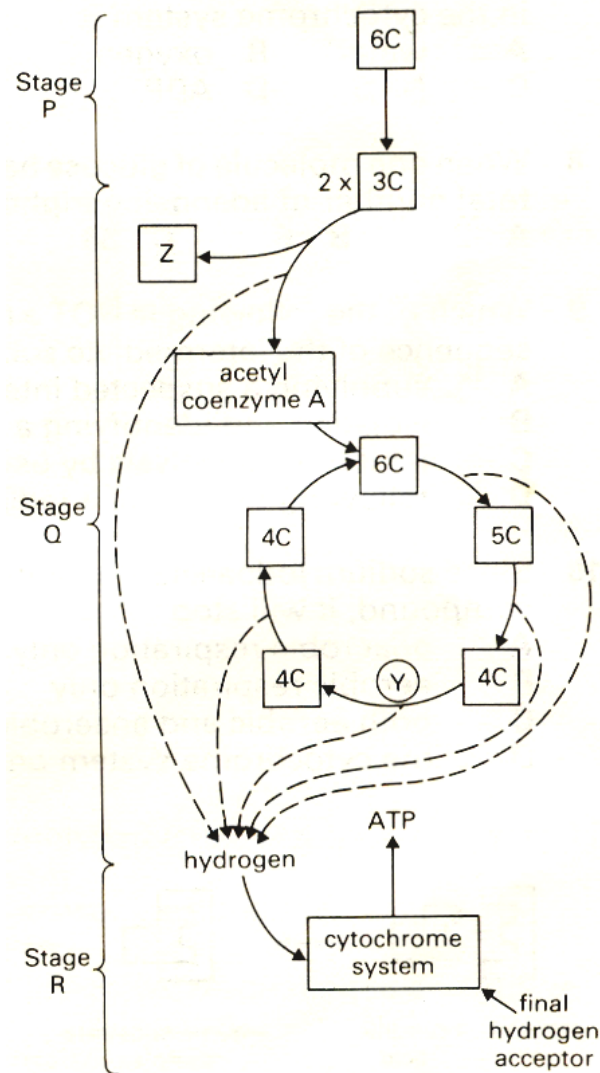
20. Which one of the following structures would be found in an animal cell undergoing mitosis, but not in a plant cell undergoing mitosis ?

A centriole
B centromere
C centrosome
D chromosome

21. Some of the photosynthetic organisms containing chloroplasts that lack PSII (photosystem II) are able to survive. The best way to detect the lack of PSII in these organisms would be to

A determine if they have thylakoids in the chloroplasts
B determine the production of starch
C test for carbon fixation in the dark
D test for the liberation of oxygen in the light

22.



Which of the following indicates the correct locations at which stages **P**, **Q** and **R** occur in a cell?

	Cristae	Cytosol	Matrix
A	R	P	Q
B	Q	P	R
C	P	R	Q
D	R	Q	P

23. The genetic material of virus has a base ratio of $(A + G) / (U + C) = 1$. What type of virus is this?

- A** a double-stranded DNA virus
- B** a double-stranded RNA virus
- C** a single-stranded DNA virus
- D** a single-stranded RNA virus

24. Which of the following statement about chromosome structure is **true**?

- 1 Linker DNA refers to the linear double-stranded DNA between adjacent nucleosomes.
- 2 The 30 nm chromatin fibre has a solenoid structure.
- 3 The 300 nm chromatin fibre is attached to multiple locations on a central protein scaffold.
- 4 Euchromatin is the more diffuse region of the interphase genetic material and is transcriptionally inactive.

- A** 1 and 2
- B** 1 and 4

- C** 2 and 3
- D** 3 and 4

25. Which of the following statement is **true** of the end-replication problem of linear DNA?

- 1 When a linear DNA molecule replicates, a gap is left at the 3' end of each new strand because DNA polymerase can only add nucleotides to a 5' end.
- 2 Telomerase prevents the end-replication problem from occurring.
- 3 Repeated rounds of replication produce shorter and shorter DNA molecules.
- 4 Prokaryotes do not have the end-replication problem.

- A** 1 and 2
- B** 1 and 4
- C** 2 and 3
- D** 3 and 4

26. The completion of the Human Genome Project revealed that the human genome contains fewer genes than expected, not so many more than simpler organisms. Which of the following statements could explain this phenomenon?
- 1 RNA transcripts of human genes are more likely to undergo alternative splicing.
 - 2 Polypeptide domains are combined in a variety of ways.
 - 3 Post-translational processing adds diversity to the resulting polypeptides.
 - 4 Gene expression patterns in humans are often more complex than those in other organisms.
- A** 1 only
B 1 and 2
C 2, 3 and 4
D 1, 2, 3 and 4
27. Which of the following statements concerning proto-oncogenes is **false**?
- A** They can code for proteins associated with cell growth.
 - B** They are produced by somatic mutations induced by carcinogenic substances.
 - C** They can be involved in producing proteins for cell adhesion.
 - D** They can code for proteins involved in cell division.
28. In *E. coli*, the production of enzymes for tryptophan synthesis is carefully controlled according to the organism's metabolic needs. A mutation in the gene encoding the tryptophan repressor has occurred, such that the repressor can bind DNA without the co-repressor. What effect on enzyme production can be expected under this condition?
- A** high-level enzyme production under any condition
 - B** high-level enzyme production in the absence of tryptophan, no activity in the presence of tryptophan
 - C** no enzyme production in the absence of tryptophan, high-level activity in the presence of tryptophan
 - D** no enzyme production under any conditions
29. In *E. coli* bacteria, regulation of gene expression involving the catabolite activator protein (CAP), is actually a type of positive regulation because
- A** cAMP-CAP helps RNA polymerase to initiate transcription.
 - B** CAP increases the production of cAMP.
 - C** glucose binds to CAP and activates it.
 - D** glucose stimulates the production of cAMP.

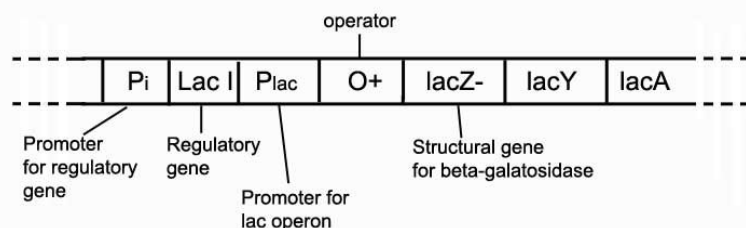
30. Some events that take place during generalized transduction are listed below

- I Bacterial host DNA is fragmented
- II Bacterial DNA may be packaged in a phage capsid
- III Recombination between donor DNA and recipient DNA
- IV Phage infects a bacterial cell
- V Phage DNA and proteins are made

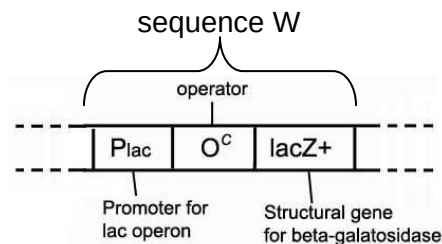
Which sequence of events is correct?

	First -----> Last
A	IV, I, III, V, II
B	IV, I, V, II, III
C	IV, III, I, V, II
D	IV, V, I, III, II

31. The diagram below depicts a *lac* operon of the wild type bacterial chromosome in *E. coli*. The *lacZ* gene is regulated via the normal operator sequence (O^+). However, the *lacZ* allele is a recessive form (*lacZ*⁻) that does not code for beta-galactosidase.



This particular *E. coli* strain is conjugated with a mutant strain of *E. coli* carrying the P_{lac} O^c *lacZ*⁺ DNA sequence as shown below. In this sequence, the *lacZ* allele is normal (*lacZ*⁺) but the operator (O) is mutated. The O^c mutant is unable to bind the *lac* repressor coded for by *LacI*.



How would the expression of the *lacZ* gene be regulated in the recipient cells after homologous recombination between the wild type (recipient) and the mutant strain (donor) involving sequence **W**?

- A** There will be normal regulation of lactose metabolism.
- B** There will be continuous expression of beta-galactosidase regardless of glucose level.
- C** The inability to synthesize the *lacZ* gene product, beta-galactosidase.
- D** The *lacZ* gene is inducible, but unable to be repressed by high glucose levels.

32. Many signal transduction pathways use second messengers to
- A** amplify the message by directly phosphorylating proteins.
 - B** induce a signal from the outside to the inside of the cell.
 - C** relay the message from the inside of the membrane throughout the cytoplasm.
 - D** transport a signal through the lipid bilayer portion of the plasma membrane.
33. Which of the following is a similarity between G-protein-linked receptors and tyrosine-kinase receptors?
- A** activation that results from binding of GTP
 - B** formation of a dimer following binding of a signal molecule
 - C** regions of the receptor that span the plasma membrane
 - D** signal-binding sites specific for steroid hormones
34. What does an increase in the secretion of insulin produce?
- A** a decrease in glucose metabolism
 - B** an increase in blood sugar level
 - C** an increase in glucose permeability of cell
 - D** an increase in the conversion of glycogen to glucose
35. Four events in the transmission of nerve impulses across synapses are:
- 1 depolarisation of the presynaptic membrane
 - 2 propagation of postsynaptic action potential
 - 3 hydrolysis of transmitter substance
 - 4 rupturing of synaptic vesicles
- In which sequence do these events occur?
- A** 1 3 2 4
 - B** 1 4 2 3
 - C** 4 1 3 2
 - D** 4 3 1 2
36. A plasmid has two antibiotic resistance genes, one for ampicillin and one for tetracycline. It is treated with a restriction enzyme that cuts in the middle of the ampicillin gene. DNA fragments containing a human globin gene were cut with the same enzyme. The plasmids and fragments are mixed, treated with ligase, and used to transform bacterial cells. Clones that have taken up the recombinant DNA are the ones that
- A** are blue and can grow on plates with both antibiotics.
 - B** can grow on plates with ampicillin but not with tetracycline.
 - C** can grow on plates with tetracycline but not with ampicillin.
 - D** cannot grow with any antibiotics.

37. Cloned cDNA of beta-haemoglobin mRNA can direct the synthesis of a complete and correct polypeptide in the bacterium *E. coli*. However, this is not so for the corresponding eukaryotic gene because

- A** bacterial polymerases cannot transcribe intervening sequences.
- B** bacteria lack the enzymes necessary for splicing eukaryotic mRNA precursors.
- C** intervening sequences contain hairpin loops that block ribosomal function.
- D** intervening sequences contain codons that are not recognized by tRNA.

38. Nine skeletons were found in an unidentified grave. To establish the relationship between these nine individuals, small amounts of DNA were isolated from these skeletons and genetic fingerprinting was then carried out.

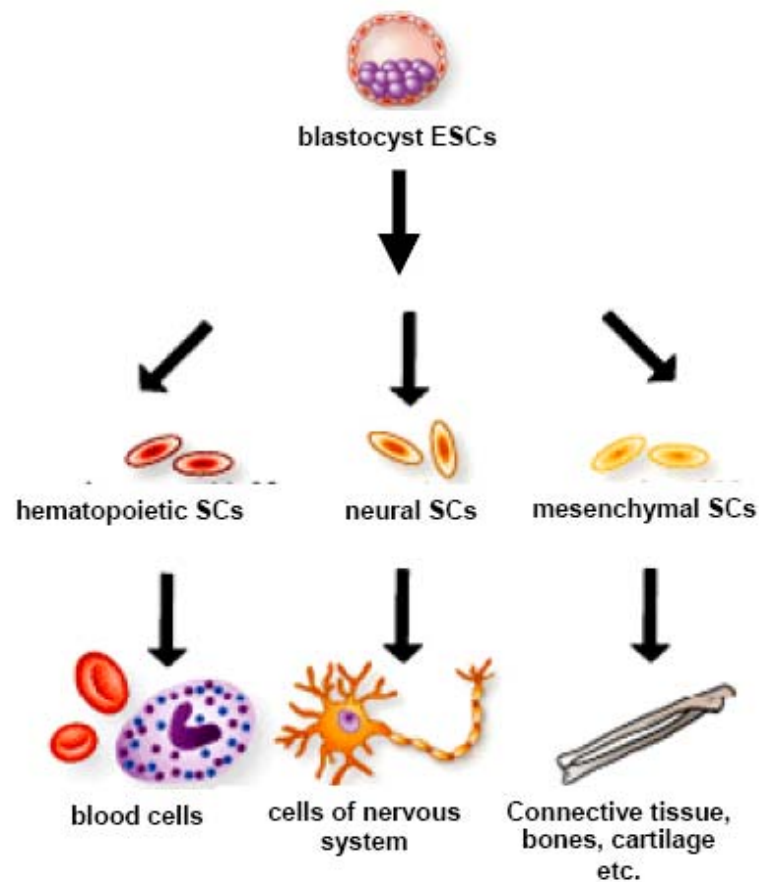
The chart shows some of the results obtained from the genetic fingerprinting of seven of the skeletons: three children and four adults.

Child 1	Child 2	Child 3	Adult 1	Adult 2	Adult 3	Adult 4
		—			—	
—	—				—	—
—	—	—	—		—	—
—	—	—	—	—		—
—	—	—	—	—	—	—
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				—		
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	—	—			—	

All three children on the chart had the same parents. Which of the following shows the children's parents?

- A** Adult 1 and 2
- B** Adult 1 and 3
- C** Adult 2 and 4
- D** Adult 2 and 3

39. Large quantities of useful products can be produced through genetic engineering involving
- A bacteria containing recombinant plasmids.
 - B mammals producing substances in their milk.
 - C transgenic plants.
 - D all of the above.
40. What feature of embryonic stem cells (ESCs) is illustrated below?



- A ESCs are capable of dividing indefinitely.
- B ESCs are multipotent.
- C ESCs are pluripotent.
- D ESCs are totipotent.

End of Paper